ENGINEERING SCIENCE

Revised SEPT 2022

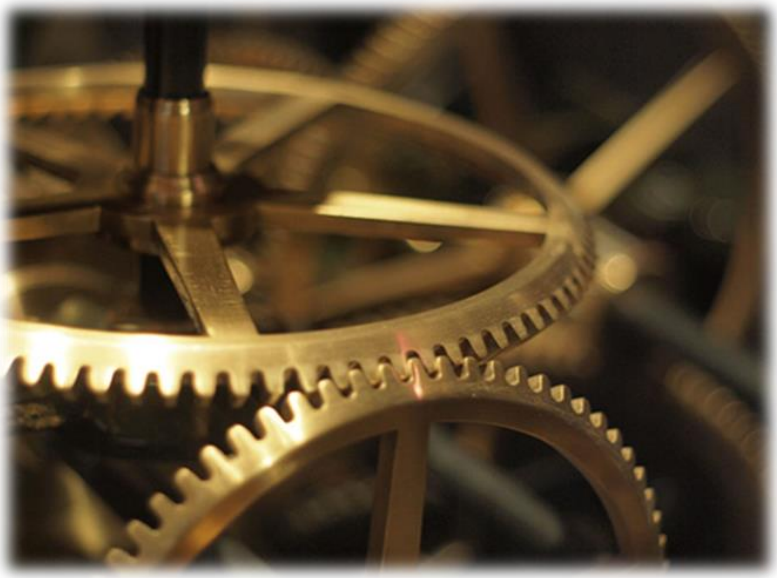
Mechanisms and Structures:

Topic 1: Types of motion and transfer of motion

Topic 2: Introduction to forces

Topic 3: Pulleys

Topic 4: Gears



Name: _____

Class: _____

Teacher: _____

Contents

Unit outline 3
Learning Intentions 3
Success Criteria 3
Mechanical symbols 4
What is a mechanism? 5
Motion 6
Rotary 6
Linear 6
Reciprocating 6
Oscillating 6
Transfer of motion 7
Static forces 7
Dynamic forces 7
Bending forces 8
Shear forces 8
Torsion forces 8
Compression forces 8
Tension force 9
Force 9
Calculating 'stress' on members due to force 10
Calculating 'strain' on members due to force 11
Converting motion (transfer in motion) 12
CAMS 12
Cam motion 12
Rotary motion to reciprocating motion 13
Rack & pinion 13
Transfer of motion 13
Couplings: Aligned and Non-aligned 15
Aligned shafts 15
Non-aligned shafts 15
Non-aligned shafts continued 16
Friction within a system 16
1. The use of lubricants 16
2. The use of bearings 16

Torque and drive systems 17
What is torque? 18
How is torque calculated? 18
Basic torque equations 18
Power transmitted by a belt drive 19
Drive systems: Smooth belt 21
Different types of smooth belt 21
Drive systems: Jockey pulley 22
Stepped cone pulley system 22
Toothed belts drive systems 22
Chain drives systems 22
Drive systems 24
Velocity ratio for belt drives 25
Reversing pulley rotation and compound system 29
Gears 30
Simple gear train 30
Movement Multiplier Ratio in gears 30
Velocity Ratio in gears 32
Idler gears 33
Simple gear trains: Worked example 33
Velocity ratio for chain drives 33
Simple gear train: Complete the following tasks in your jotter 34
Different gears and drive systems 35
Worm and Wheel 35
Ratchet and pawl 35
Worm and nut 35
Bevel gears 35
Chain tension 35
Compound gears 37
Compound gear train: Complete the following tasks in your jotter 38
Exam style questions 40

Unit outline

The general aim of this unit is to develop an understanding of mechanisms and structures. Learners will explore a range of mechanical and pneumatic systems and design, simulate, construct, test and evaluate mechanical or pneumatic solutions to solve problems.

Outcome 1

The learner will:

1 Investigate a range of mechanical and pneumatic systems by:

1.1 Describing or producing diagrams of a range of structures.

1.2 Describing or producing diagrams of a range of pneumatic systems.

1.3 Describing or producing diagrams of a range of mechanical drive systems.

1.4 Carrying out calculations involving energy, work, power and efficiency using given formulae.

Outcome 2

The learner will:

2 Develop mechanical or pneumatic solutions to solve problems by:

2.1 Identifying key aspects of a problem.

2.2 Applying knowledge and understanding of structures, pneumatics and/or mechanical drive systems.

2.3 Designing, and simulating or building, mechanical or pneumatic systems.

2.4 Testing and evaluating solutions against a specification.

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red:	I need to work on this topic further to understand it.
Amber:	I need a bit of help to fully understand this topic.
Green:	I am confident in my understanding of this topic.

Learning Intentions

- To understand basic gear and drive systems.
- To get a basic knowledge of the workings of a range of simple everyday engineered objects.
- To gain knowledge of key facts related to mechanical systems.
- To know how mechanisms can create motion transfer.
- Be able to explain the examples of motion in mechanical systems.
- To know the purpose of an idler gear and jockey wheel.

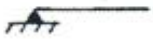
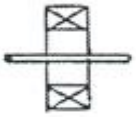
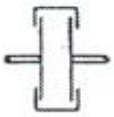
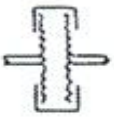
Success Criteria

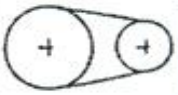
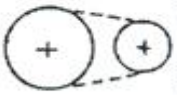
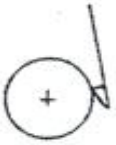
☐ I am able to apply knowledge and understanding of key engineering facts and ideas.

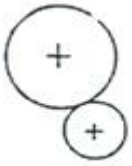
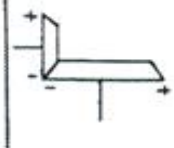
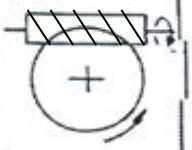
☐ I am able to use knowledge of mechanisms and drive systems in a practical means.

☐ I can explain the following terms using diagrams to help: linear, reciprocating, rotary and oscillating.

Mechanical symbols

DESCRIPTION	SYMBOL
MOVEMENT	
Linear	
Rotary	
Linear (reciprocating)	
Rotational (reciprocating)	
Fulcrum	
Lever and Fulcrum	
Shaft	
Ball or Roller Bearing	
Coupling	
Flexible Coupling	
Cam	

DESCRIPTION	SYMBOL
Belt Drive	
Chain Drive	
Flywheel	
Governor	
Ratchet	
Pump	
Compression Spring	
Tension Spring	

DESCRIPTION	SYMBOL
Torsion Spring	
Torsion Spring	
Spur Gears	
Spur Gears (+ Movement towards observer)	
Bevel Gears	
Worm and Wheel	
Rack and Pinion	

What is a mechanism?

Mechanisms are widely used in industry and are included in a large number of devices and equipment commonly used in society. It is important, therefore, that students have a good understanding of how mechanical systems assist and aid everyday tasks. Throughout the coursework, students should gain knowledge of how mechanical systems operate and how they fit into common everyday devices. It is not only necessary to understand how the mechanical system carries out its task, but also important to know the systems can be altered to improve performance and efficiency.

Why use a mechanism?

Mechanisms are essential in manipulating energy within moving systems. Mechanisms allow us to drive cars, use machines and even tasks as simple as open doors. The World we live in is surrounded by mechanisms so it is important that as Engineers we fully understand the place that mechanisms have in a modern society.

Mechanisms are devices that will convert one form or size of motion into another, or will convert one kind or size of force into another. A mechanism is a system of moving parts that changes an input motion and force into a desired output motion and force. Mechanisms are a large part of modern society. Most of the mechanisms are that we use everyday are so familiar that we never think twice about them, for example door handles, light switches, scissors.

Mechanisms play a vital role in industry. While many industrial processes now have electronic control systems, it is still mechanisms that provide the muscle to do the work. They provided the forces to press steel sheets into the shape of car body panels, to lift large components from place to place and to force power hacksaws to cut through thick metal bars – the list of jobs is endless. It is only by using mechanisms that industry can make products you use every day.







Some machines are easy to understand, but many are hidden away from sight behind glossy panels and covers. In the past, machines were much easier to see, as with the old steam engine, for example, but as people became more concerned about safety, it was necessary to fit guards over moving parts. Today, guards are often replaced by styled covers that make it much harder to see what is happening, but whether you can see them or not, mechanisms are still playing a vital part in everyday life.



A Mechanism changes an input motion and force into another output motion and force. All mechanisms must:

- involve some kind of motion.
- involve some kind of force.
- make a job easier to do.
- need some kind of input to make them work.
- Produce some kind of output.

Motion

To understand how a mechanism works, it is vital you know the ways it can move. There are four basic kinds of motion:

Rotary



Turning in a circle. This is the most common type of movement, for example wheels, clock hands, CD-ROMs, compact discs or gears. Do NOT identify 'Rotary' motion as 'rotational'.

Linear





Movement in a straight line, for example movement of a paper trimmer cutting a straight edge on paper or a lift moving between floors.

Reciprocating





Backwards and forwards movement in a straight line, for example the needle in a sewing machine or the piston in a car engine.

Oscillating





Swinging backwards and forwards in an arc, for example the pendulum of a clock, a playground swing or a rocking horse.

1. ***Motion:***

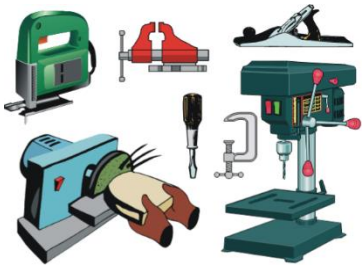
In your jotter, identify what type of motion are the following scenarios represent:

Rotary, Linear, Reciprocating or Oscillating

- A person running a 100 metre race:
- Car wheels turning:
- A child on a swing:
- A person swinging a golf club:
- A person cutting wood with a saw:

Transfer of motion

It is possible to use mechanisms to transfer motion. This means one type of motion can be the input to a system and another type of motion can be the output.



2. The machines and tools that are used in your practical rooms in school use all types of motion. The four types of motion are listed; now list as many machines / tools as possible under each heading of motion. Answer in your jotter.

Rotary Linear Reciprocating Oscillating

Forces

Forces affect structures in a variety of different ways depending on how they are applied to the structure. Forces can move a structure slightly or cause damage by changing the shape of the structure.

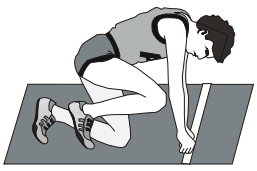
Sometimes when forces are applied to a structure, it may be almost impossible to see changes happening. For example, a bridge will sag slightly when a vehicle drives over it, but this is not visible to the human eye. Nevertheless, the vehicle causes downwards movement of the bridge structure. Loads such as vehicles on a bridge can be deemed examples of forces acting on the bridge. Forces can stop an object from moving or they can make it change direction. When a football is kicked, the forces applied from the player cause the dimensions of the ball to change on impact. It happens so quickly that it is not visible.

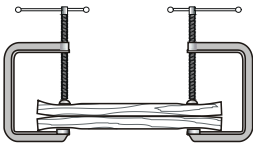
Forces are measured in *Newtons* and the symbol is the letter '**N**'.

There are a number of different types of forces that can be applied to and which affect bodies and structures.

Static forces

When static loads or forces are applied to structures, the structures do not normally move. Normally the total downwards force comprises the weight of the structure plus the load it is carrying. The runner below is in his starting position; his weight is a static or stationary downwards force.





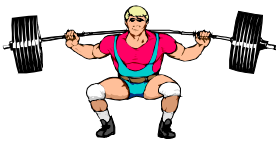
A G-Clamp is a piece of workshop equipment when gluing timber together and pressure is required.

Dynamic forces

When dynamic loads or forces are applied to a structure, the structure does move and the forces applied can be varied. Dynamic forces are visually more noticeable and are produced by a variety of means and effects: machines, wind directions, people, etc. The picture below shows the sprinter after the starting gun has been fired; he is creating a dynamic impact to gain momentum.

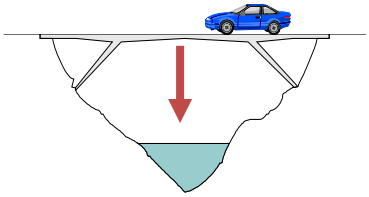


Bending forces

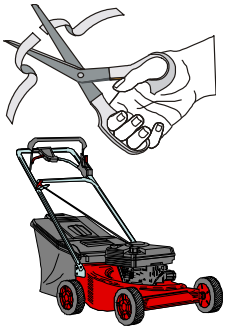


Structures that carry loads across their length are subject to bending forces. The weightlifter lifting a weighted bar feels the effect of the downward forces of the weights and these cause the bar to bend.

A car driving across a bridge will cause bending forces on the structure but often they are not visible.

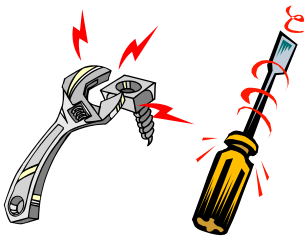


Shear forces



Shear forces can be described as tearing or cutting forces affecting structures. Simple examples are a pair of scissors used to cut a ribbon at an opening ceremony and a mower cutting the grass.

Torsion forces



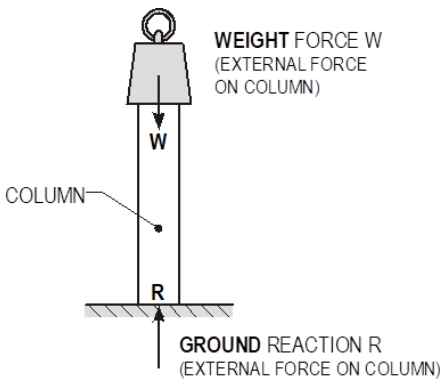
Torsion or torque forces have the effect of trying to turn or twist a structure or a piece of material. A screwdriver being twisted to apply a force to a screw and a spanner turning a bolt to lock it into place are examples of torque being applied.

Compression forces



The picture below shows the supporting columns on the new Queensferry crossing which cross over the River Forth between Fife and Edinburgh. The column experiences a huge downward force from the cables but they don't not disappear into the ground because the ground exerts an upwards reaction force on the column's base. The downward pressure of the weight and the upward reaction are external forces trying to squash or shorten the column. Forces that act like this are called compressive forces and the column is said to be in compression. The Queensferry Crossing is called a *cable stayed* bridge design.



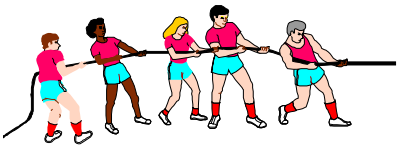


For example, when you sit on a stool in the classroom, your weight acts as a downward force on the chair. However, there must be an upward force on the legs of the chair; therefore the legs are said to be in compression. The same can be said about the weightlifter's arms and legs.



Tension force

We have noted that compression occurs when things are being pushed together. The opposite of compression is 'tension' – when a structure is being pulled apart. In a tug of war, the two sides are pulling the rope in opposite directions. The forces applied by the teams are called tensile forces and cause the rope to be in tension. It could also be said that the arms of team members are in tension.





The wire rope holding the net in volleyball is also in tension.



Force

3. Against each of the seven forces mentioned, make a list of 'real life' situations where these types of forces may be found. You may research the internet to provide some examples. Answers in jotter.

a) Static

b) Dynamic

c) Bending

d) Shear

e) Torsion

f) Compression

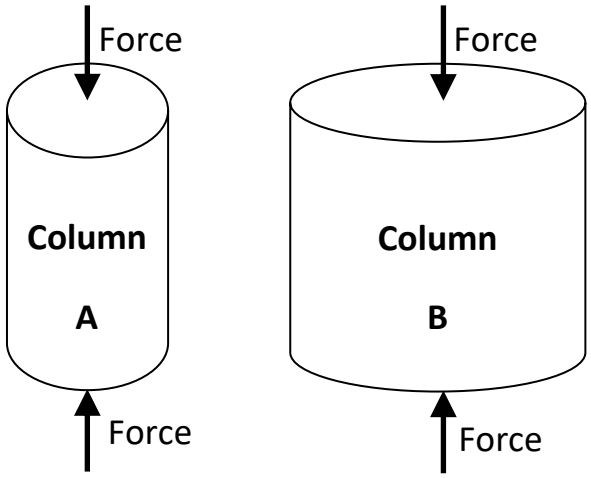
g) Tension

Calculating 'stress' on members due to force

When mechanical or structural engineers are working on a project, they need to select the right materials (wood, steel, iron, plastic etc). There are three things they need to know about the structure and materials they choose to ensure it will be strong enough and whether they will stretch, compress or balance when forces are applied. When a force is applied to a part of a structure, it will experience **STRESS**. If the level of force which is applied to the member is too high then the stress which is created may cause the part to bend, buck or worst case, fail and break.

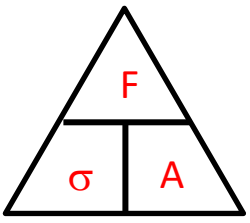
When a force is applied to a part of a structure, it will experience **STRESS**. If the level of force which is applied to the member is too high then the stress which is created may cause the part to bend, buckle or worst case, fail and break.

The stress can vary by the force put on it, but also by the size of the structure. For example; if the same load is applied to column A and column B which are of the same length, then Column A will suffer **more** from stress and will be more likely to fail. The reason why is because Column A has a far smaller cross-sectional area.



Where Force is measured in Newtons (N) and the Area is the cross-sectional area measured in mm^2. Stress therefore is measured in Nmm^{-2} (the unit measurement can also be given as N/mm^2). This means the amount of force acting on a structure is Newtons '**per**' square mm). Stress is denoted by the Greek letter Sigma (σ). It is calculated using the formula:

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} \qquad \sigma = \frac{F}{A}$$



Teacher worked stress examples on Powerpoint

4. In your jotter, copy the follow questions and answer.

A. A rectangular bar of 180mm x 120mm cross section is subjected to a tensile load of 500N. Calculate the stress in the bar.
B. A wire 4.0mm in diameter is subjected to a force of 300N. Calculate the stress of the wire.
C. The stress in a steel wire supporting a load of 8.0kN should not exceed $200N/mm^2$. Calculate the minimum diameter of wire required to support the load.

D. A column of cross section 0.73m^2 is required to act as a roof support. The maximum allowable working stress in the column is 4.0 MNm^{-2}. Calculate the maximum compressive load acting in the column.

E. The figure shows an oil platform used in the North Sea. The platform is designed so that each of the three legs always carries an equal share of the **120 MN** load. The legs are **15.0m** in diameter and are made of steel reinforced concrete. Calculate the stress that is applied on **each leg.**

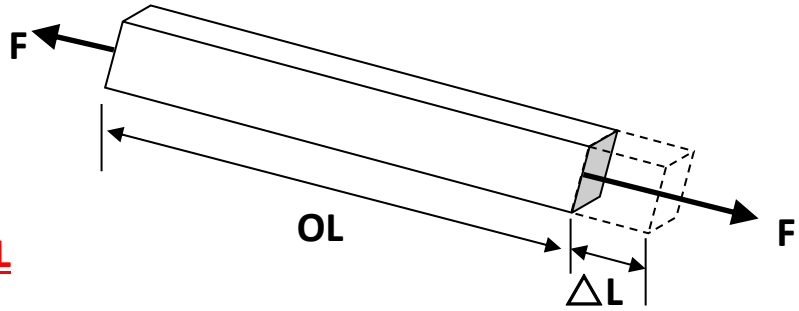


Calculating 'strain' on members due to force

The result of applying a load or force to a structural member is a change in length. Every material changes shape to some extent when a force is applied to it. This is sometimes difficult to see in materials such as concrete and we need special equipment to detect these changes.

If a *compressive load* is applied to a structural member, then the length will reduce. If a *tensile load* is applied, then the length will increase. The result of applying a load to a structural member in **either** direction is called **STRAIN**. The △L (Delta L) means the '*change*' in length when compared against the original OL.

Strain is calculated using the formula:



$$\text{Strain} = \frac{\text{Change in Length}}{\text{Original Length}} \qquad \varepsilon = \frac{\triangle L}{OL}$$

Where length in both cases is measured in the same units (m or mm). As the units cancel each other out, **strain is dimensionless**. This means that there are **no units of strain**. Put simply, strain is a ration that describes the proportional change in length in the structural member when a direct load is applied. Strain is denoted by the Greek letter *Epsilon* (**ε**).

Teacher worked strain examples on Powerpoint

5. In your jotter, copy the follow questions and answer.

a) A steel wire 4.0m long is used to support a load. When the load is applied the wire stretches by 2.5mm. Calculate the strain of the wire.

b) A wire 10m long stretches 5.0mm when a force is applied at one end. Calculate the strain.

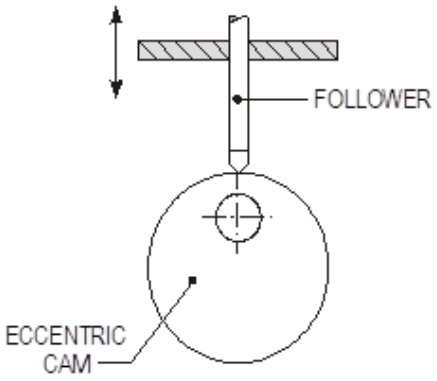
c) The allowable strain on a bar is 0.0075 and its length is 2.5m. Calculate the change in length.

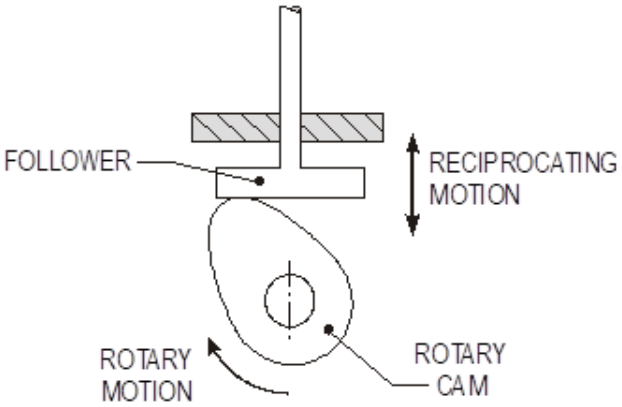
d) During testing, a steel rod stretches 0.6mm. If the resulting strain was 0.0012, what is the original length of the rod?

Converting motion (transfer in motion)
CAMS

A cam is a specifically shaped piece of material, which can be used to change an input rotary motion to an output motion that is oscillating or reciprocating.

The cam operates by guiding the motion of a follower held against the cam, either by its own weight or by a spring. As the cam rotates the follower moves. The way that it moves and the distance it moves depends on the cam's shape and dimensions.

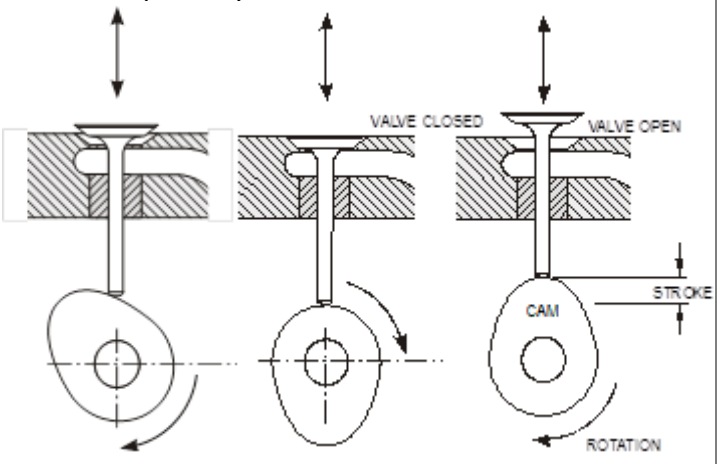


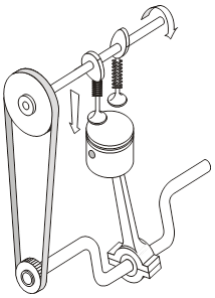


Cam motion

Pear-shaped cams are often used for controlling valves. For example they are often used on motor-car cam shafts to operate the engine valves.

A follower controlled by a pear-shaped cam remains motionless for about half a revolution; during the other half revolution of the cam the follower rises and falls. As the pear-shaped cam is symmetrical, the rising motion is the same as the falling motion. When the follower is not moving we call this the dwell part of the cam.

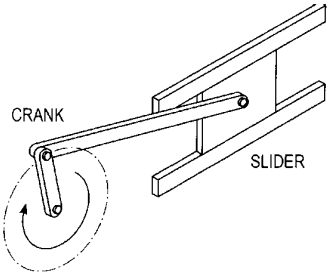




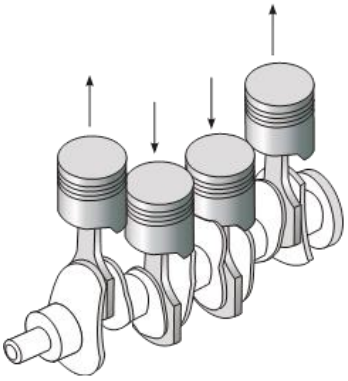
In a car engine, cams are fixed on a camshaft. As each cylinder has two valves, an inlet and an exhaust valve, there are two cams on a camshaft for each cylinder as shown.

Crank & slider

Crank & slider mechanisms involve changes between rotary and reciprocating motion. The crank rotates while the slider reciprocates. The longer the crank the further the slider will move.

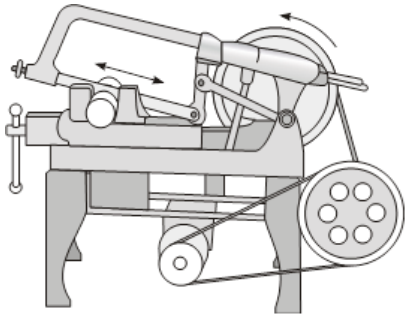


Reciprocating motion to rotary motion



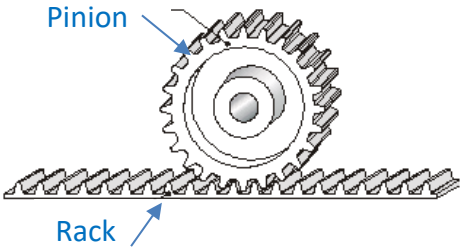
Car engines use reciprocating pistons, which are connected to a crankshaft by connecting rods. As the piston moves up and down the connecting rods push the crankshaft round. Each piston moves down in turn so keeping the crankshaft turning.

Rotary motion to reciprocating motion



A power hacksaw uses an electric motor to power a crank, which is connected to a saw frame. The saw frame is free to slide on the arm. As the crank rotates it causes the frame to slide backwards and forwards on the arm. The longer the crank the further the saw frame will move.

Rack & pinion



A rack & pinion mechanism is used to transform rotary motion into linear motion, or linear motion into rotary motion. A round spur gear, the pinion, meshes with a rack that can be thought of as a spur gear with teeth set in a straight line.

Gear wheels are normally made from metal or plastic. Plastic gears have the advantage that they are much quieter running and need less lubrication. Plastic gears however wear down much faster. Consider the gearing within a cordless drill.

Transfer of motion

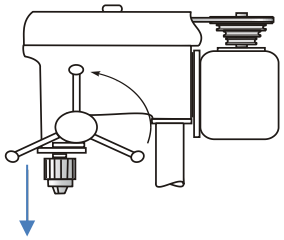
It is possible to use mechanisms to transfer motion. This means one type of motion can be the input to a system and another type of motion can be the output.

6. Using the website technologystudent.com and your course notes, sketch the mechanical image of the transfer of motion as a 2D graphic (examples are on page 4 of this booklet) and create brief notes on the following mechanisms in your jotter. Your sketches must be labelled, axel rotation cross hair positions and you should explain what type of motion is converted from one form to another.

- Cam and follower. (there are 4 shapes of CAM's you must identify and explain their unique/difference follower action.
- Crank, link and slider.
- Rack and Pinion.
- Chain and Sprocket. (Requires your own research)

7. Think about where the mechanisms in question 6 could be used in real life. In your jotter, write down some examples for each type of mechanism.

8. A rack on a pillar drill with 100 teeth per 10 cm is meshed with a pinion that has 10 teeth. Answer in your jotter.



a) If the pinion rotates one revolution, how far does the rack move?

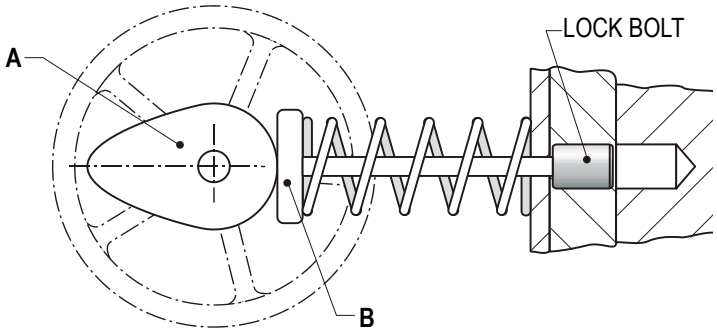
b) How many revolutions does it take to move the rack from one end to the other?

c) What is the change in motion for this system?

d) Where would a gear system like this be used?

9. Past paper questions. You do not need to copy the questions into your jotter.

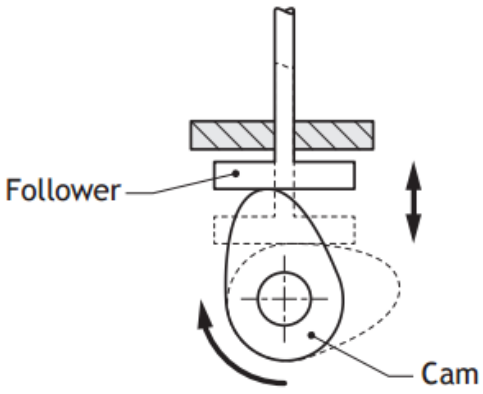
The pear-shaped cam and follower shown represent a simple locking mechanism.



Name parts A and B

A _____ B _____

The cam and follower in the diagram below are used to convert motion.



State the type of **motion** of:

the cam; _____

the follower. _____

Couplings: Aligned and Non-aligned

Rotary machines employ a variety of methods of transmitting motion from one part of a machine to another. The motion is often transmitted through shafts, which are round metal rods. Often these shafts must be connected together to transmit the motion.

Shafts are joined using a device called a coupling. In small models, such as those used in schools, simple sleeves or tubes of plastic use friction to drive two shafts, which are pressed into the sleeve. Stronger couplings are required for industrial-sized machines.

Aligned shafts

Where shafts are in line with each other they are joined either with a flanged coupling or a muff coupling. All couplings must be 'keyed' to the shafts they are joining to give a positive drive. Figure 1 shows a flange coupling then a muff coupling in figure 1(b).

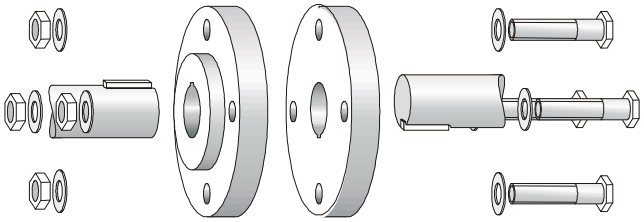


Figure 1(a): flange coupling

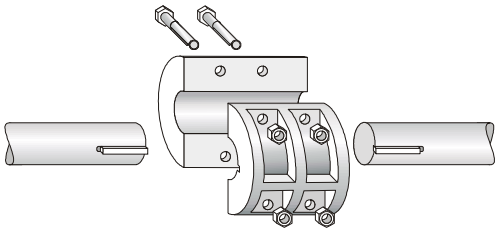
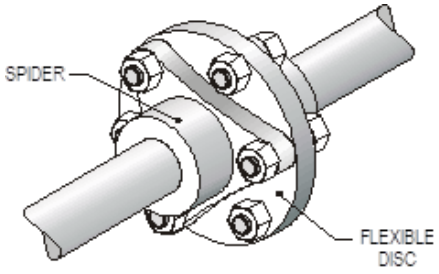


Figure 1(b): muff coupling

Non-aligned shafts

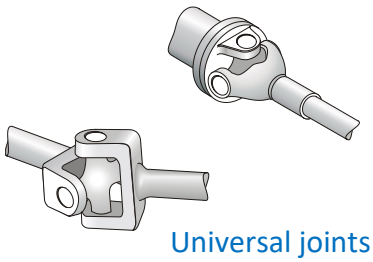
Where shafts meet at a slight angle, some method of compensating for misalignment must be used. Where the misalignment is small, a flexible coupling (flexi-coupling), using either rubber or a mixture of rubber and steel, is used. The rubber is flexible enough to compensate for small changes in angle (below).



flexi-coupling

Non-aligned shafts continued

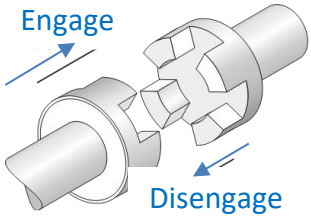
When the alignment is more than a few degrees out, a universal joint is used. A universal joint can transmit motion through an angle of 20 degrees. Figure on the right is a Hooke's universal joint. The two yokes are free to pivot on the central 'spider'. Modern universal joints use needle roller-bearings between the spider and the yokes.



Universal joints

Clutches

Clutches are devices that allow two rotating shafts to be connected and disconnected. There are two types of clutch, the positive clutch and the friction clutch. A dog clutch is a positive clutch and has four interlocking block on one shaft that can be interlocked with 4 on the other shaft. A simple friction clutch relies on two plates to transmit the power from one shaft to another.



Friction within a system

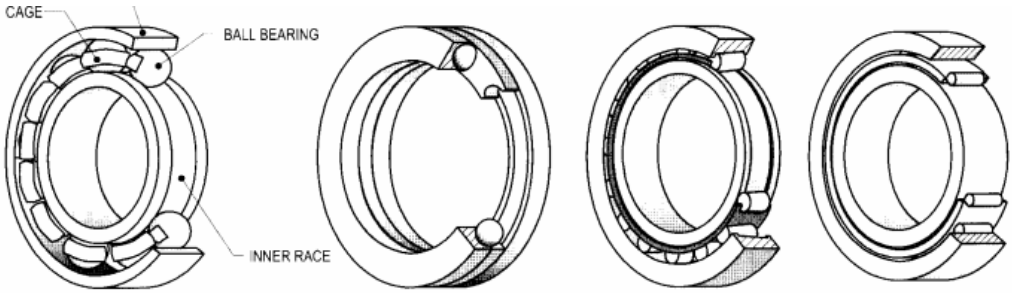
Within all drive systems, a huge amount of friction can be created, causing heat. This can cause the material of the gears/pulleys to expand, which can cause a lot of damage to the drive system. To help reduce this, two things can be done:

1. The use of lubricants

By lubricating the moving parts with grease or oil, it allows the different parts to touch and move more easily without causing so much heat. If you reduce heat due to friction then you can also increase efficiency and reduce noise from parts rubbing together (cause and effect).

2. The use of bearings

These are components that are design to withstand wear. Ball and roller bearings are used within high-speed and high-force systems. Bearing replace any rubbing action with a rolling action instead as rolling does not create as much friction as rubbing. Parts of mechanisms that slide over each other use flat bearings, which can be made from brass or bronze and replaced when they are worn. Split bearings are designed to wear, it stands to reason that they must be able to be removed and replaced. Grease can be used within bearing but only within sealed systems where dirt will not get in.



Ball Thrust ball Roller Needle roller

10. Explain several ways of reducing friction within a mechanism. Explain using cause and effect.

Torque and drive systems

A force is needed, but it must be applied in a way that creates a twisting or turning action. Torque, T, is the rotational version of force, and results from the application of one or more forces. Torque is specified relative to a chosen rotation axis or pivot.

Torque is dependent upon:

- The distance from the rotation axis to the force application point (refer to figure 1 below).
- The magnitude of the force, **F**.
- The orientation of the force relative to the displacement from the axis to force application point (refer to figure 2).

Anyone who has ever opened a door has an intuitive understanding of torque. When a person opens a door, they push on the side of the door **farthest** from the hinges. Pushing on the side **closest** to the hinges requires considerably more force. Although the work done is the same in both cases (the larger force would be applied over a smaller distance) people generally prefer to apply less force, hence the usual location of the door handle.

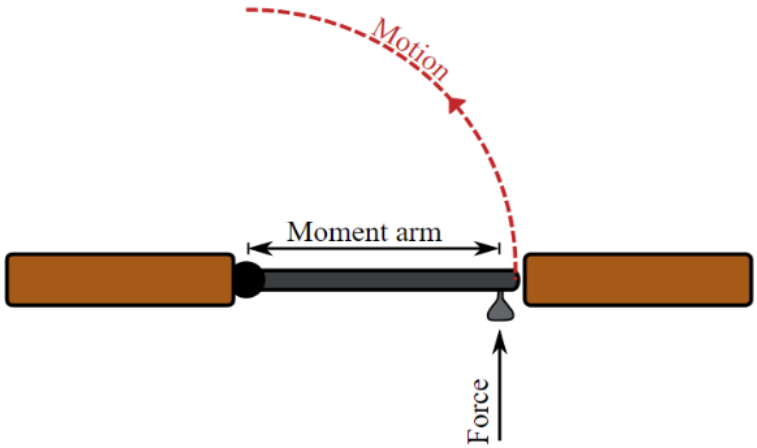


Figure 1: Opening door with maximum torque

There are two things which we use to measure 'Work'.

Work = Force x distance

We already know that a force is measured in Newtons (N), and distance is measured in metres (m), therefore the unit will be Newton-metres (Nm). This unit of work is also given the special name, Joule (J).

For example: 1 Nm or 1 Joule of '*Work*' is generated when a Force of 1 Newton moves the body on which it is acting 1 metre in the direction of the force.

What is torque?

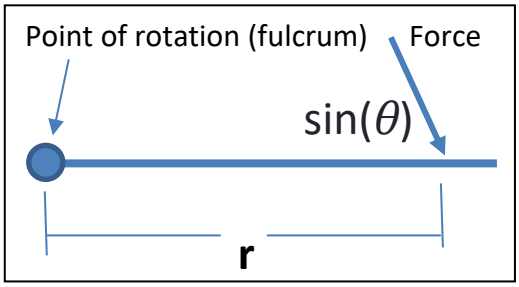
In simple terms, the definition of torque is the engine's '**rotational force**'. It differs from horsepower as it refers to the amount of work an engine can exert, while horsepower defines how quickly that work can be delivered. This is why torque is often referred to in layman's terms as 'pulling power', 'oomph', or 'grunt'.

For example, a Ford Fiesta Ecoboost car develops peak torque between 1,400 rev/min and 4,000 rev/min. Generally speaking, a car will feel more responsive when peak torque is developed low down in the rev range, but plenty of people enjoy high-strung engines that need to be revved hard before they develop peak torque. Torque becomes more important as vehicles get larger and heavier. A tiny city car can easily cruise along with very little torque, where as a big SUV or panel van requires a lot to get it moving. This is why larger vehicles tend to be fitted with diesel engines - diesels offer more torque at lower revs than petrol engines.

How is torque calculated?

The magnitude of the torque vector T for a torque produced by a given force F is

$$T = F \cdot r \sin(\vartheta)$$



where 'r' is the length of the moment arm and ϑ is the angle between the force vector and the moment arm. In the case of the door shown in Figure 1, the force is at right angles (90°) to the moment arm, so the Sin ϑ term becomes 1 and simplifies the equation to:

$$T = \text{Force x radius } (T = F.r)$$

Basic torque equations

Let's say we're using a 0.5m long wrench to tighten a wheel nut, and we need to lean on the far end of the wrench with a force of 50 Newtons to do it up tightly. Simply multiplying the two numbers gives us the required torque figure in Newton metres:



Torque = Force x distance (r)

Torque = 50 (N) x 0.5 (m)

Torque = 25Nm or 25J

Teacher worked example of Torque on Powerpoint

Copy this question into your jotter.

How much torque is required to tighten the nut if the force required is 45N and the radius of the tool is 200 mm?

Answer on Powerpoint presentation

Teacher worked example of Torque on Powerpoint

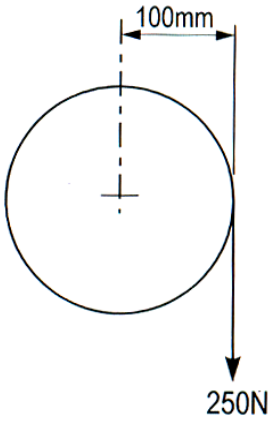
In your jotter, copy the follow question and answer.

How much torque is required to tighten the nut if the force required is 45N and the radius of the tool is 200 mm?

Answer on Powerpoint presentation

Teacher worked example of Work done

A belt drives a pulley, shown opposite, with a shaft of diameter 'd' of 200mm. If the effective belt tension tending to turn the pulley is 250 N, calculate the ***Work done per revolution****.*



Answer on Powerpoint presentation

Power transmitted by a belt drive

Teacher worked example of power transfer

The effective pull on a belt drive is 420 N when driving a 500 mm diameter pulley. The speed of the rotation is 220 revolutions ***per minute****. Calculate the Power (Watts W).*

Power (/s) = circumference ($2\pi r$) × 'n' is the rotational speed per second (/s) x Torque (Force x radius)

Work done = 2πnT

Note the units of measurements:

When a Force 'F' (Newtons), acts at the rim of a pulley '***r***' *(radius and metres), 'n' is the rotational speed* per second (/s). Calculating Power **must be done per second** and is given by the formula:

Power = 2πnT

The effective driving force 'T' = Force x radius

$= (T_1 - T_2)$. r

T_1 is on the tight side and T_2 is the tension on the slack side. But there is only **1** force given in the question, therefore (T_1) = 420 N

Remember to change to its proper units so the final answer is in the correct units. Change diameter 'd' of 500mm to d = 0.5m but it must be a radius. Therefore r = 0.25m

Number of revolutions '**n**' per second (/s) = 220 ÷ 60 = 3.67 revolutions per second

Therefore

r = 0.25m

T_1 = 420N

n = 3.67 revs/sec or 3.67 revs sec^{-1}

Therefore Power transmitted = $2\pi n$ ('effective driving force' (T_1-T_2) . r)

= $2\pi n (T_1-T_2)$. r

= 2 x π x 3.67 x (420) x 0.25

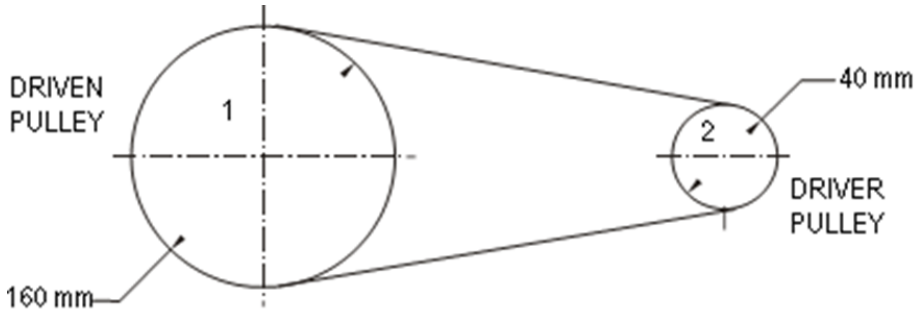
= 2421.23 **Watts** = 2.42 **kW**

= 2.4KW (2 s.f.)

Drive systems: Smooth belt

Many mechanisms make use of rotary motion, often provided by someone turning a handle or by an electric motor. But to be useful, this rotary motion has to be frequently transmitted from one part of a mechanism to another, often with a change of speed by varying the diameter of the driver pulley and the driven pulley. While gears can be connected together in a simple gear train, if too many gears are connected then it can result in large efficiency losses through friction.

There are two simple means of transmitting rotary motion over relatively large distances; belt or chain. The tensioned belt transmits the rotary motion from pulley 2 to pulley 1 but also a change of speed is achieved. The belt is tightened or tensioned by pulling one of the pulleys out and locking it in place. Pulleys are thin metal discs with a groove cut into the circumference of the disc.

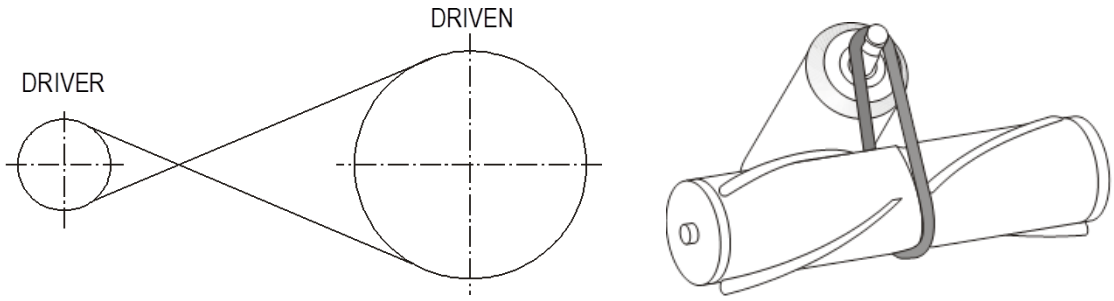


Different types of smooth belt

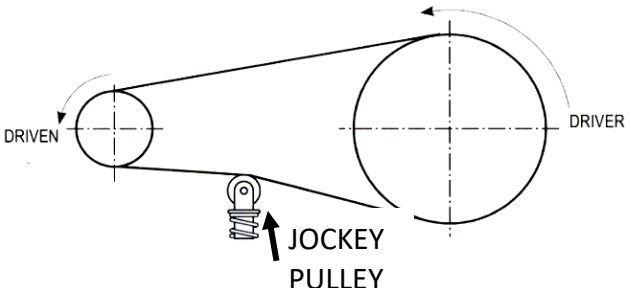
Belt drives are used in a wide variety of situations. They are made from a composite of two materials, rubber and string. The string helps to prevent the rubber from stretching too much. Drive belts are inexpensive to produce. They are easy to replace and need little maintenance, as they do not require lubrication. Belt drive systems are found in many household machines such as washing machine, tumble driers and vaccum cleaners.



A ‘V’ or ‘Vee’ belt is angled to give better grip which prevents the belt from slipping when compared against a flat belt design. A change in direction can also be achieved if the belt is crossed over as shown below. V-belts can also be used to transfer motion 90 Degree of the driven from the driver.

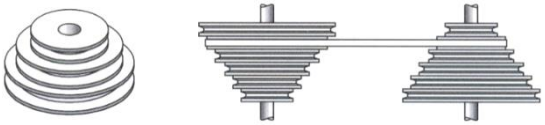


Drive systems: Jockey pulley



The advantage of a belt drive system is that it will absorb shock as it is **allowed** to slip. It is used in machines where you would not want the motor to stop or cease. For instance, if a belt drive is used to transmit the power from a motorcycle engine to the rear wheel and the biker tries to 'wheelie', the smooth belt allows slippage, preventing damage to the engine. However, excessive slipping will create inefficiency in the system. At the same time, if the belt is too tight the pulley bearings will get damaged. One way of keeping the correct tension is to use a spring loaded jockey pulley as shown above.

Stepped cone pulley system

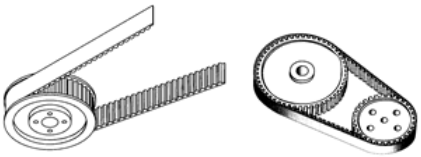


A variety of output speeds and torques can be achieved using this system. The drive motor is attached to one set of pulleys and the drive belt can be moved between various pairs of pulleys to give a selection of speeds.

Toothed belts drive systems



Toothed belt

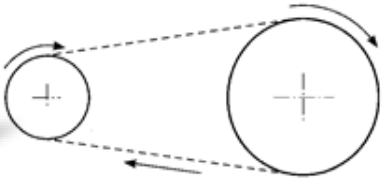


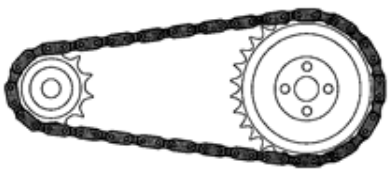
Belt drives tend to use their ability to slip to their advantage, however where slippage would damage a mechanism toothed belts have been developed that retain the advantages of normal belts but **do not slip**. Many cars have toothed belts (for example timing belts) to control the opening and closing of the inlet and outlet valves in the car engine. If the belt slipped, the pistons would collide with the valves, damaging the engine. These belt drives are quiet, require little maintenance and are easily changed if required

Chain drives systems

Where large forces have to be transmitted and there can be **no slippage allowed**, chain drives are used. Instead of a pulley, a toothed wheel known as a sprocket is used to drive a chain. Different sprockets sizes will vary speed. Can also be called a Chain and Sprocket 'transfer of motion'.

Notice: Chain system diagrams are drawn with **dashed** lines.





The best example of a chain drive system is the one found on bicycles. When cycling, if you want to go suddenly faster, you stand up and put extra weight into the pedals, which will in turn transfer greater forces to the back wheel and make you go faster or easier. Chain systems transfer greater forces as they are much stronger, unless the chain or sprockets is damaged. However, they do have to be oiled regularly, and both the chain and sprockets are prone to wear. They are also more expensive and much noisier than belt drives.

Calculating the output speed of a simple pulley

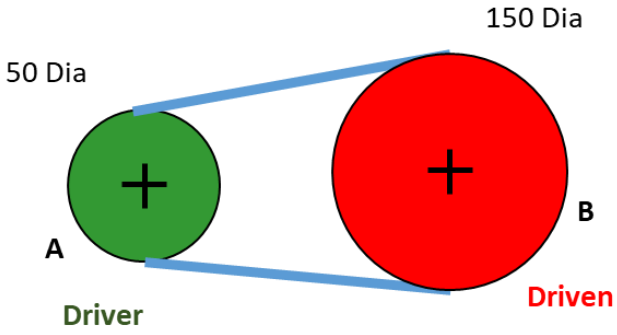
Knowing the diameter of each pulley and the input speed, then it is easy to calculate the output speed. Doing calculations with belt drives is very similar to gears. The simplest method is to use the following equation:

Diameter x TURNS = Diameter x TURNS

(Driver) (Driven)

Teacher worked example on Powerpoint: Copy these questions into your jotter

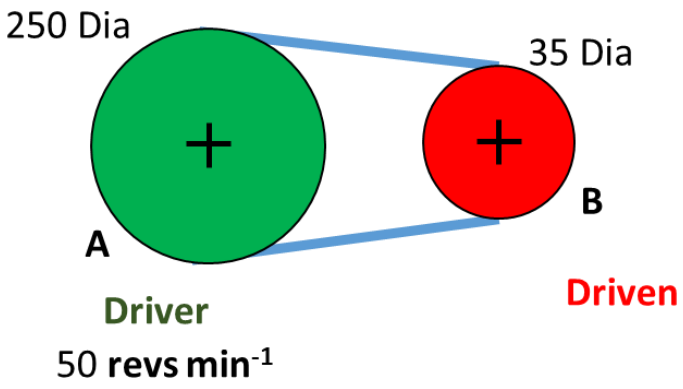
Driven to go **slower**:



10 **revs min^{-1}**

Diameter (pulley A) x speed (of pulley A) = Diameter (pulley B) x speed (of pulley B)

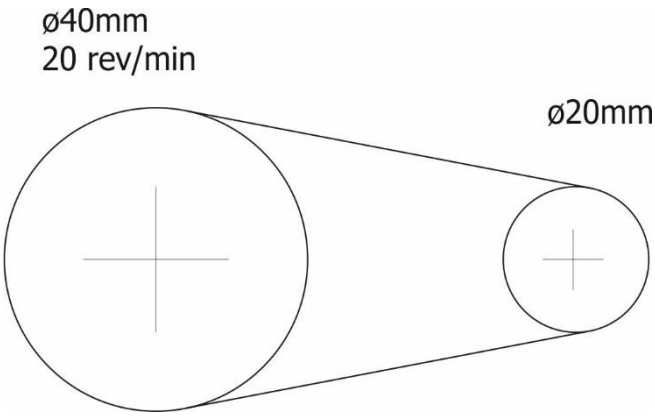
Driven to go **faster**:



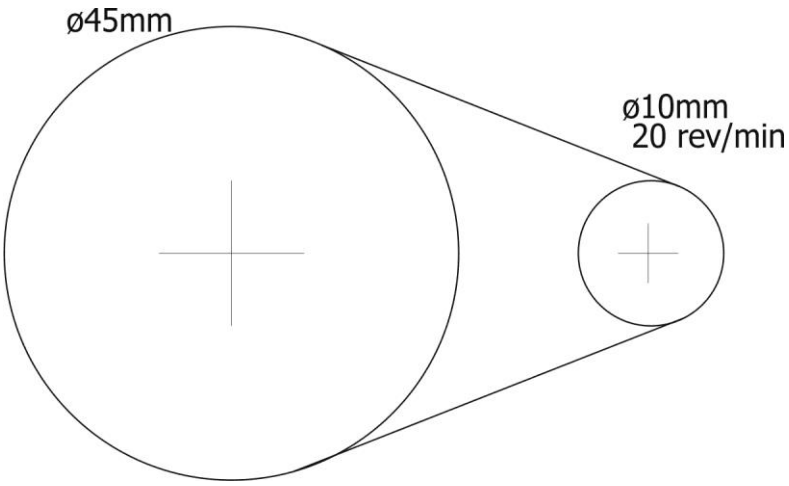
Drive systems

11. Copy the following belt drive systems into your jotter and calculate the missing piece of information. It might be to calculate the revolutions per minute or the diameter of the wheel.

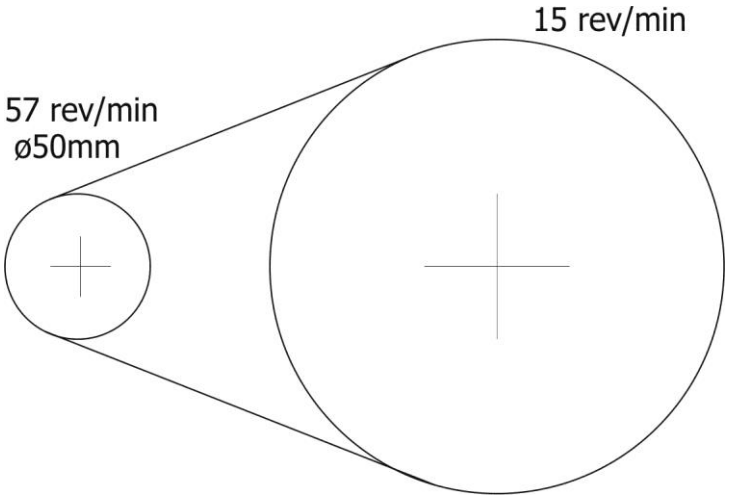
a)



b)



c)

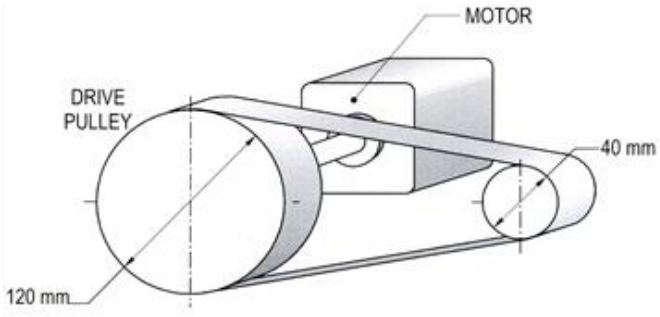


d) Draw a symbol for two pulleys that produce a decrease in speed and with a change in direction for the driven pulley.

Velocity ratio for belt drives

Pulley systems can be used to transmit rotary motion over a large distance. The input rotary motion is often from a fixed speed and fixed torque electric motor. Changing the ratio of the diameters of the pulleys can very both the speed of the output and the torque at the output.

This motor is connected to a pulley of 120mm ø. This is the driver pulley. The driven pulley is 40mm ø. To work out the Velocity Ratio for a pulley system, use this equation (**belt system**):

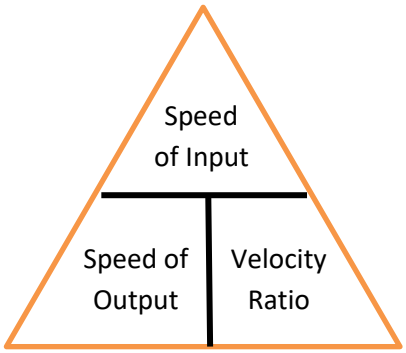


$$\textbf{Velocity ratio} = \frac{\textbf{diameter of driven pulley}}{\textbf{diameter of driver pulley}}$$

For the system in the diagram above, the Velocity Ratio is: VR = 40 ÷ 120 = (0.333) or 1 : 3

However, when calculating **speed,** use this equation:

VR = Speed of Input / Speed of Output (provided in data booklet)



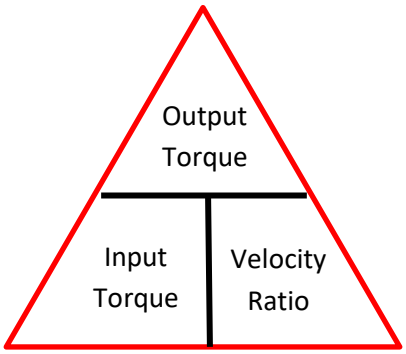
Therefore re-arranging the equation:

Speed of Output = Speed of Input / Velocity Ratio

Teacher example

In your jotter, copy the question and answer.

The washing machine motor has a top speed of 8000 rev / min. The machine has a velocity ratio of 6:1. Calculate the speed of the output drum.



However when calculating **torque**, it is a different equation:

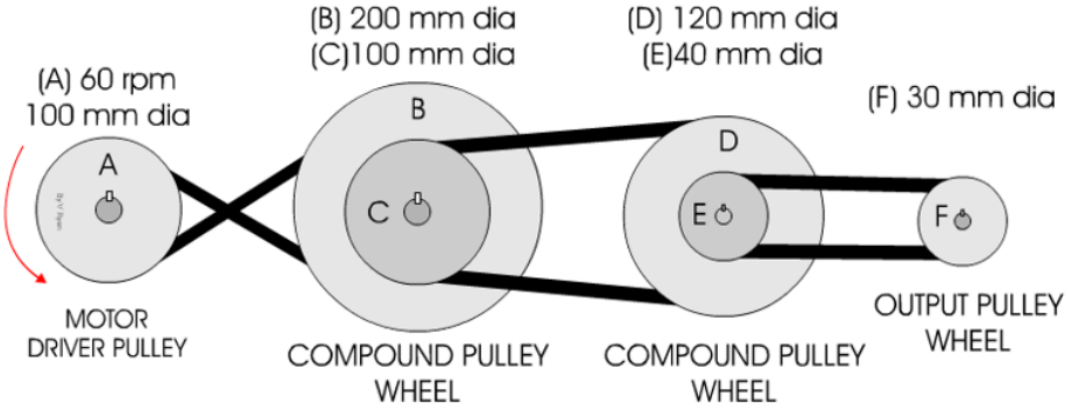
As speed increases, torque reduces but as speed reduces, torque increases.

Therefore: **VR for a 'turning force/torque' = Output torque ÷ Input torque**

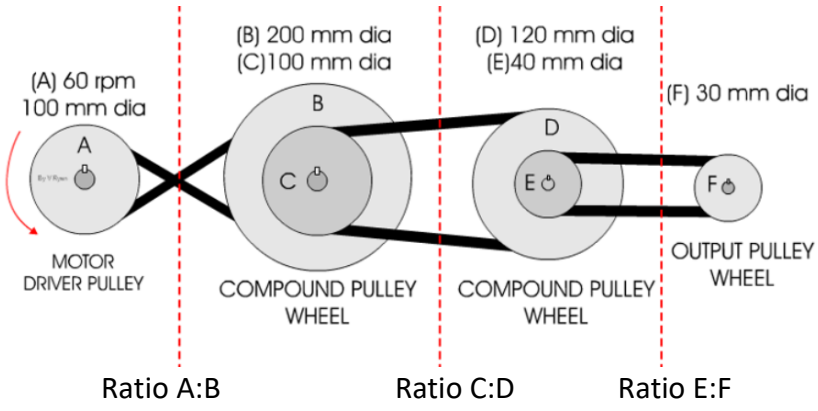
Teacher worked example: Question from 'Technology student' and printable worksheet.

A system of four pulley wheels are set up as shown in the diagram below. The driver pulley rotates in an anti-clockwise direction.

a) What direction does the output pulley wheel revolve?
b) Calculate the final output speed of pulley F



To answer the question, spit the pulleys into pairs and work out the velocity ration of each pair. Treat the pairs of pulleys as separate questions. Note that pulley's B and C are on the same axel to the ratio does NOT change.



Velocity ratio = $\frac{\text{diameter of driven pulley}}{\text{diameter of driver pulley}}$

Calculate the Velocity Ratio between pulleys A : B

Calculate the Velocity Ratio between pulleys C : D

Calculate the Velocity Ratio between pulleys E : F

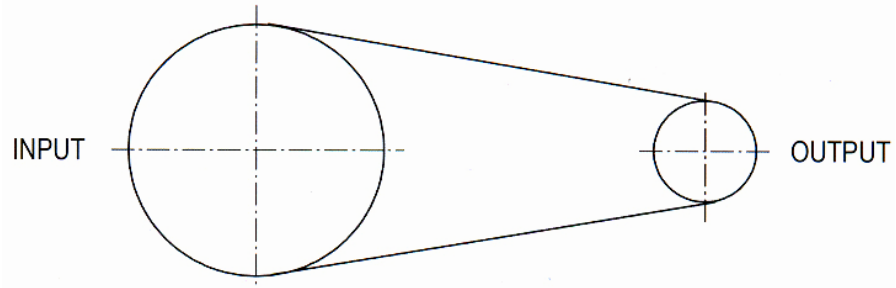
Pulley wheel A has an revs / min of 60. Pulley B is larger and so revolves at a lower rate than A. This means that the revs / min of A is divided by the ratio.

Pulley wheel C has the same revs / min as pulley B because they form a compound pulley. Pulley D is larger and so revolves at a lower rate than C. This means that the revs / min of C is divided by the ratio

Pulley wheel E has the same revs / min as pulley D because they form a compound pulley. Pulley F is smaller and so revolves at a higher rate than E. This means that the revs / min of E is divided by the ratio

12. a) Copy into your jotter and label the line diagram of the belt showing:

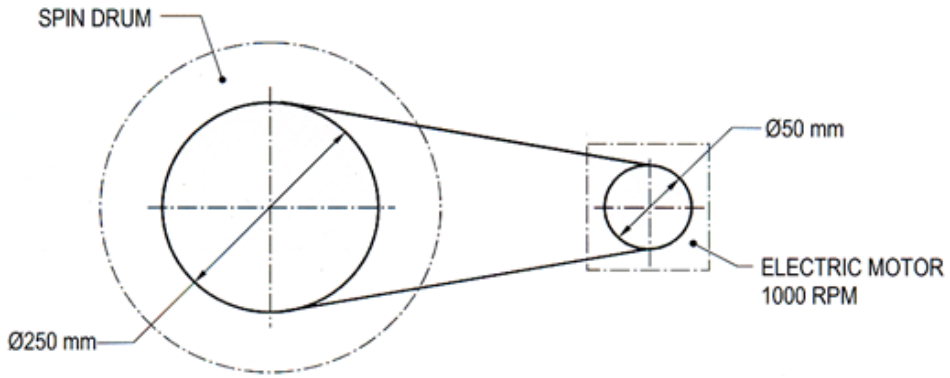
- The driver pulley
- The driven pulley
- The belt



b) In the system above, when the driver is turned, does the driven pulley turn faster or slower than the driver?

c) If the diameter of the driver pulley is 40mm and the diameter of the driven pulley is 10mm, what is the velocity ratio?

13. This is a belt drive system for a washing MACHING DRUM. It is for transmitting rotary motion from an electric motor to a spin dryer system. The motor produces a force of 800N at 1000 rev / min.

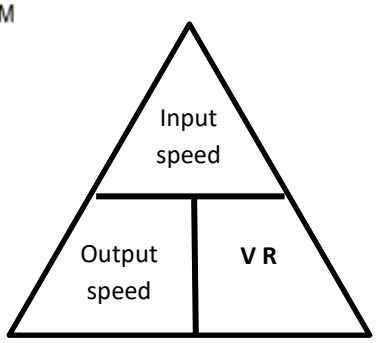


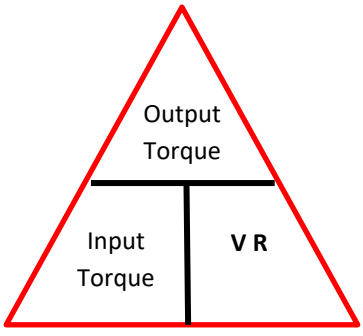
In your jotters, calculate:

a) The velocity Ratio of the system

b) The **output speed** of the drum

c) The **output torque** produced at the drum. Remember the motor produces a 'force' of 800N going into the drum. The equation formula changes



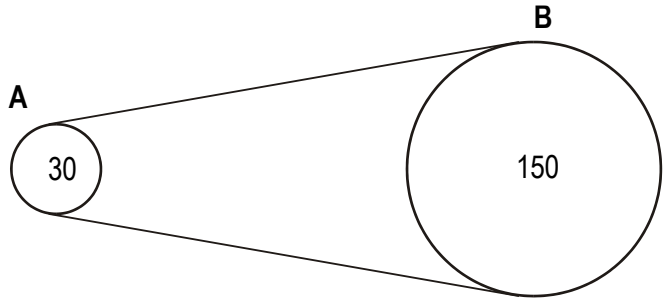


Reversing pulley rotation and compound system

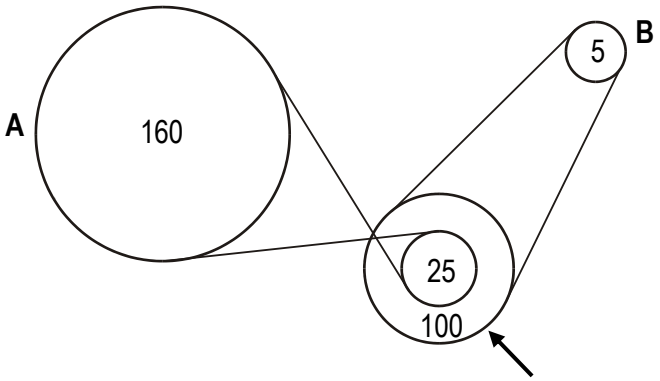
14. In your jotter, calculate the **velocity ratios** for the following belt-drive systems. To answer the question, spit the pulleys into pairs and work out the velocity ration of each pair. Treat the pairs of pulleys as separate questions. They are driven from A to B. Also indicate with an arrow the direction of rotation of B assuming A is clockwise.

Velocity ratio = $\frac{\text{diameter of driven pulley}}{\text{diameter of driver pulley}}$

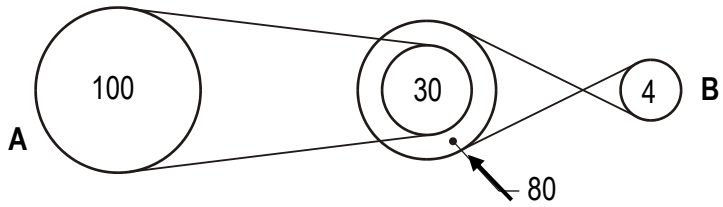
A)



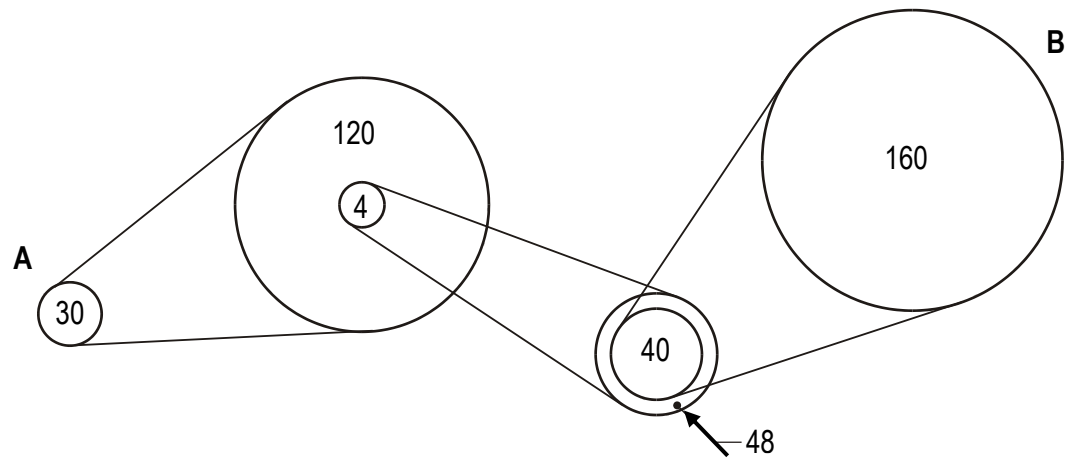
B)



C)



D)

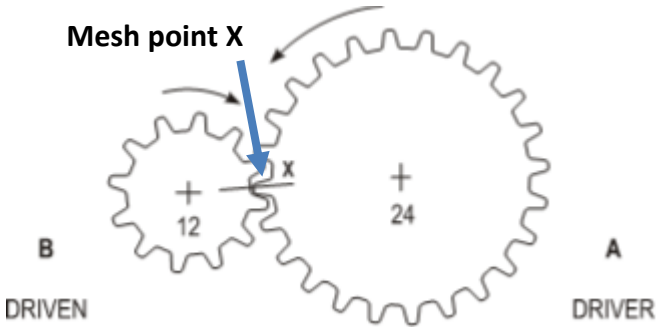


Gears

As you know, gears are toothed wheels designed to transmit rotary motion and power from one part of a mechanism to another. They are fitted to shafts with special devices called **keys** (or splines) that ensure that the gear and the shaft rotate together. Gears are used to increase or decrease the output speed of a mechanism and can also be used to change the direction of motion of the output.

Simple gear train

Gears work by interlocking or meshing the teeth of the gears together as shown. When 2 or more gears are **meshed** they form a **gear train**. The input gear which causes the system to move is called the **driver** and the output gear is called the **driven**.



Movement Multiplier Ratio in gears

The ratio of change in speeds between the gears used to be called the **Movement Multiplier Ratio (MMR)** but **now known** as the **Velocity Ratio (VR)**. The ratio of a gear system is found by dividing the number of teeth on the driven gear by the number of teeth on the driver gear. Using the simple gear train diagram above, if driver gear A has 24 teeth and it makes one complete turn, then 24 teeth of gear B must pass point **X** on the diagram. If driven gear B is meshed with driver gear A, then for every tooth of gear A to pass point **X**, one tooth of gear B must pass this point. Therefore, to be able to do this, gear B must make two complete revolutions but in the opposite direction.

We can calculate the **Velocity Ratio** with gears very easily using the following equation;

$$\textit{TEETH x TURNS} = \textit{TEETH x TURNS}$$

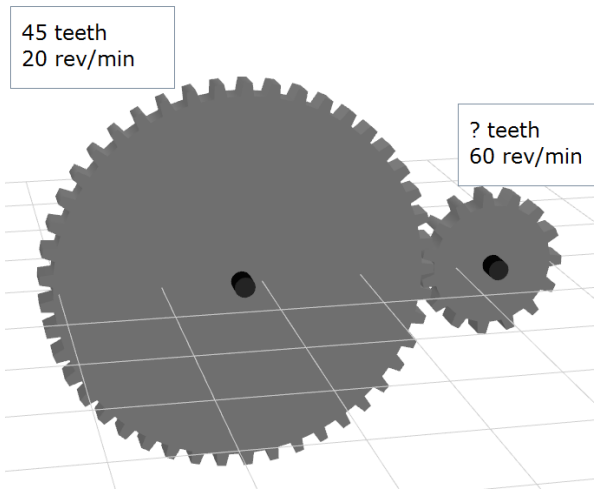
(Driver) (Driven)

15. Work out the missing quantity (speed or number of teeth) in the following gear trains. Use Yenka to build the gear trains and check your answers. Design two other simple gear train systems and print a copy of the working system beside your calculations in your jotter.

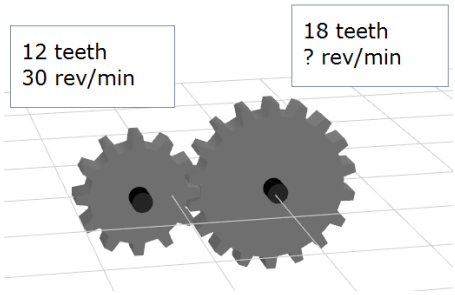
Q: Why do I need to know how print a simulation from Yenka?

A: You might need to print an example of gear simulation for the final assignment.

a)

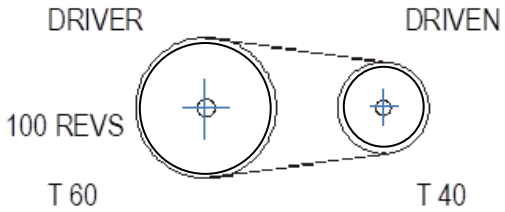


b)

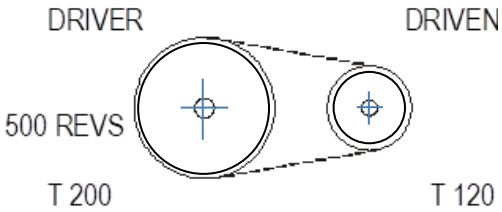


16. Four gear drive systems are shown below. In your jotter, copy the diagram and calculate the output speed for each. They are linked by a chain (dashed line).

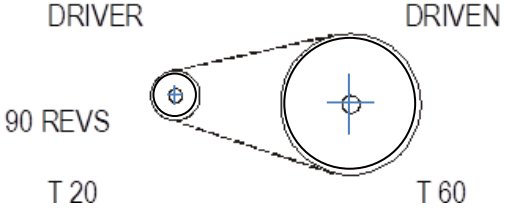
a



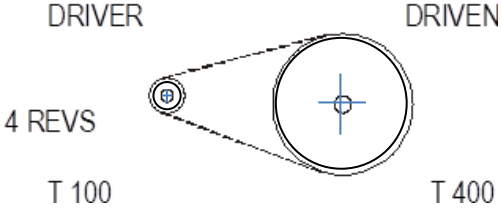
b



c



d



Velocity Ratio in gears

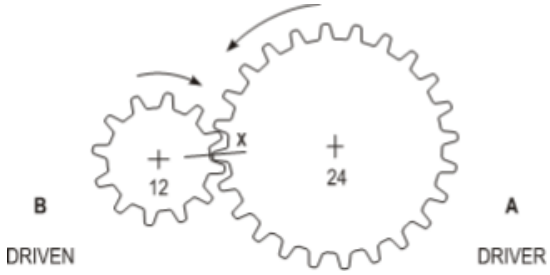
The ratio of change in speed between the gears is called the Velocity Ratio (VR).

The ratio of a gear system is found by dividing the number of teeth on the driven gear by the number of teeth on the driver gear. This can be used to calculate the output speed of a gear system.

$$VR = \frac{\text{number of teeth on OUTPUT (driven) gear}}{\text{number of teeth on INPUT (driver) gear}}$$

Teacher worked example: Copy this question into your jotter

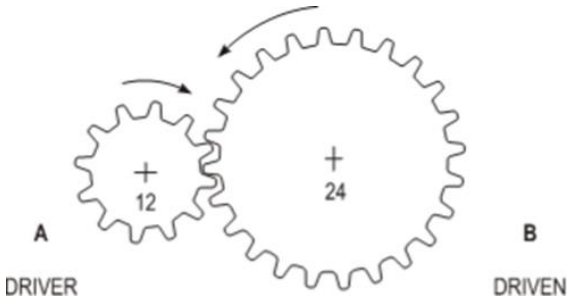
For the gear system shown opposite, the Velocity Ratio is



(The Driven will be going at **TWICE** the speed of the driver)

This means that if gear **A** was rotating at 100 rev/min clockwise then gear **B** would rotate at 200 rev/min anti-clockwise.

Gears can also be used to **decrease** the speed of a mechanism.



(The Driven will be going at **HALF** the speed of the driver)

Remember, a slowing system will have a velocity ration less than 1 and an increasing system will have a velocity ratio greater than 1.

17. You know how to build a simple gear train that will increase the speed of rotation of the driven gear compared to the driver gear. From a selection of four gear wheels: 8 t, 16 t, 24 t and 40 t, design and build a simple gear train on *Yenka* that will provide the biggest **increase** in speed between the driver and driven gears.

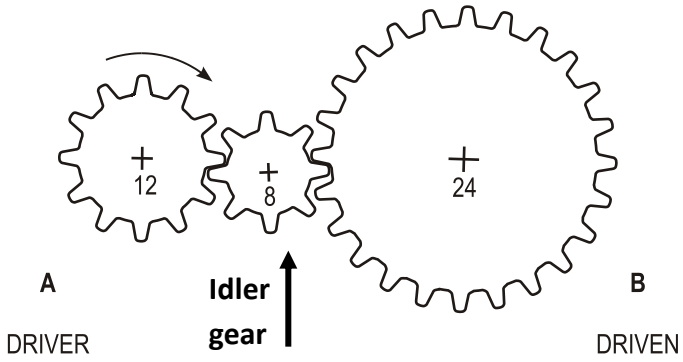
Calculate the **velocity ratio** of your system in your jotter.

18. Modify your simple gear train so that it will give you the biggest **decrease** in speed between the driver and driven gears.

Calculate the **velocity ratio** of your system in your jotter.

Idler gears

To get the driven gear to rotate in the same direction as the driver a third gear is inserted in the middle of the system. The idler gear has **no effect** on the speed of the driven gear wheel.



Simple gear trains: Worked example

Using the simple gear train shown above and if the driver rotates at 100 rev/min in a clockwise direction, calculate…

a) The Velocity Ratio.

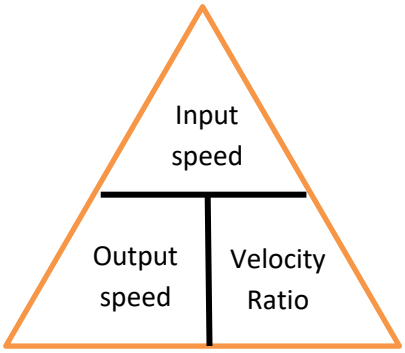
$$VR = \frac{\text{number of teeth on OUTPUT (driven) gear}}{\text{number of teeth on INPUT (driver) gear}}$$

The velocity ratio for the simple gear train shown above is still 2:1. If gear A still rotates at 100 rev/min clockwise then the output of gear B will rotate at 50 rev/min clockwise.

b) The output speed of the system.

$$\text{Output Speed} = \frac{\text{Input Speed}}{\text{VR}}$$

$$\text{Output Speed} = \frac{100}{2}$$



c) The direction of rotation of the driven gear.

As the Driver is going clockwise, the idler will turn anti-clockwise which will return the Driven as clockwise.

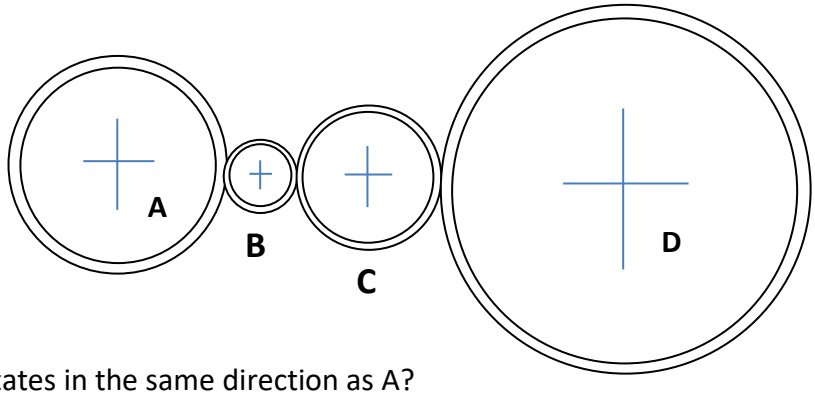
Velocity ratio for chain drives

Calculating the velocity ratio, output speed and torque of a chain drive is very similar to calculating them in belt drive systems. The calculation is:

$$\textbf{Velocity Ratio} = \frac{\textbf{no. of teeth on driven sprocket}}{\textbf{no. of teeth on driver sprocket}}$$

19. In your jotter for this simple gear train, calculate the following:

A = 50 teeth
B = 10 teeth
C = 25 teeth
D = 100 teeth



a) Which gear rotates in the same direction as A?

b) The velocity ratio of A to B, A to C and at D

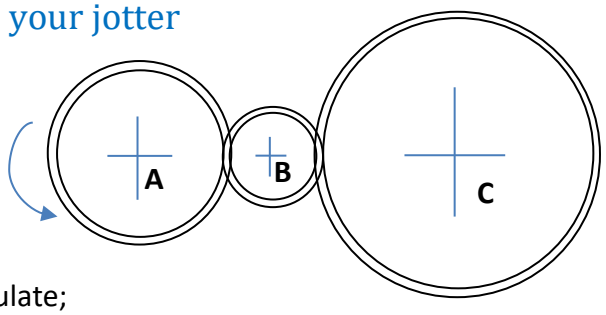
A to B A to C A to D

c) The speed of gear B, C and D if A rotates at 500 rev/min.

B = C= D=

Simple gear train: Complete the following tasks in your jotter

20. A simple gear train is shown.



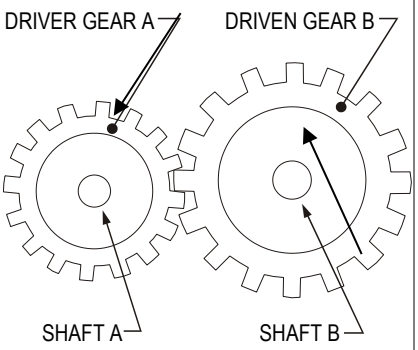
A = 20 teeth
B = 5 teeth
C = 30 teeth

If gear A rotates at 250 rev/min in a clockwise direction, calculate;

(a) The Velocity Ratio of the gear system.

(b) The output speed.

21. A simple gear train is shown. The driver gear A has 20 teeth. When shaft A is rotated 10 times, shaft B rotates five times.



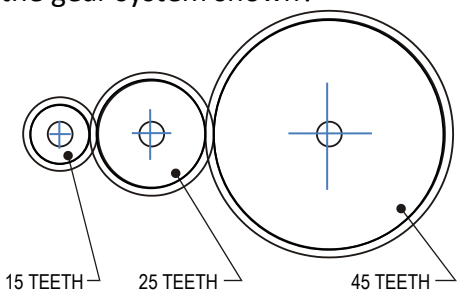
(a) How many teeth has gear B?

(b) What is the gear ratio of the system?

(c) If shaft A rotates at 60 rev/min, at what speed does shaft B rotate?

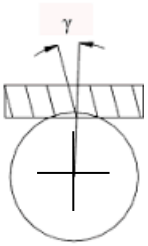
(d) If shaft A rotates anti-clockwise, in which direction does shaft B rotate?

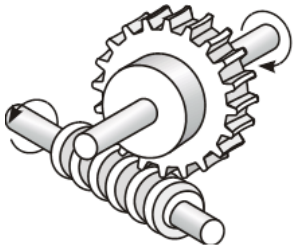
22. What is the velocity ratio of the gear system shown?



Different gears and drive systems

Worm and Wheel





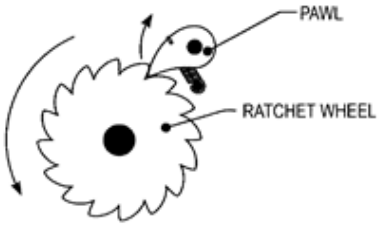
Another way of making **larger** speed reductions is to use a worm and wheel. The worm, which looks rather like a screw thread, is fixed to the driver shaft (sometimes directly onto the motor shaft). It meshes with a worm wheel, which is fixed to the driven shaft. The driven shaft runs at 90 ° to the driver shaft.



You should think of the worm wheel as a gear with only 1 tooth wrapped around a cylinder. This allows a huge reduction in speed but a huge increase in torque while taking up little space.

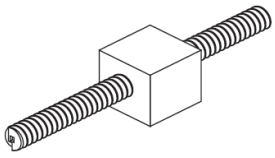
Worm wheel example
Teeth on driven: 20
Teeth on driver: 1

Ratchet and pawl



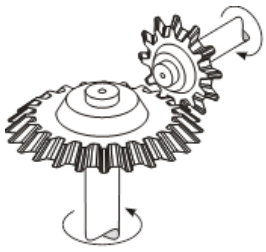
A wheel with saw shaped teeth round its rim is called a **ratchet.** The ratchet wheel usually engages with a tooth shaped lever called a **pawl**. The purpose of the pawl is to allow rotation in one direction only and prevent rotation in the opposite direction.

Worm and nut



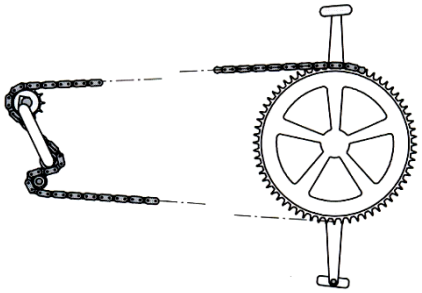
The worm gear is fixed so that when it spins, it moves the block. This transmits the motion through the gear. This allows for a big change in speed and increased torque.

Bevel gears



Bevel gears, like worm wheels, use shafts at 90° to each other. A kitchen whisk uses bevel gears to change the direction motion through 90 ° as does the gears in a wind turbine.

Chain tension

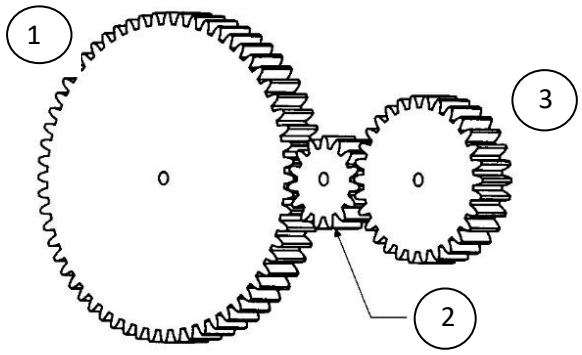


Chain drive systems must also have a means to tension the chain. If the chain is over-tensioned there will be extensive wear on the chain, sprockets and bearings in the system. To give a greater control, a spring loaded jockey wheel, such as that used in Derailleur gears on mountain bikes is used.

23. In your jotter, copy the table and sketch the mechanical symbols for the following:

Simple gear train		**Compound system of 4 gears**	
Cam		**Follower**	
Belt drive		**Chain drive**	
Rack and pinion		**Bevel gears**	

24. A simple gear train used in a food processor is shown.

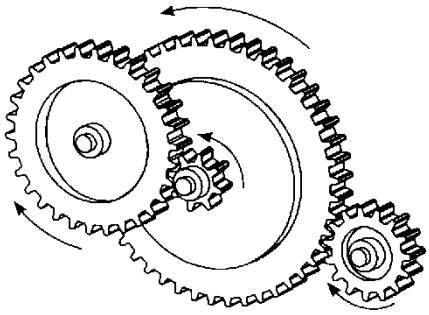


(a) Draw, using the correct symbols, this simple gear train.

Gear 2 in the simple gear train is an idler.

(b) Explain the function of an idler gear.

Compound gears

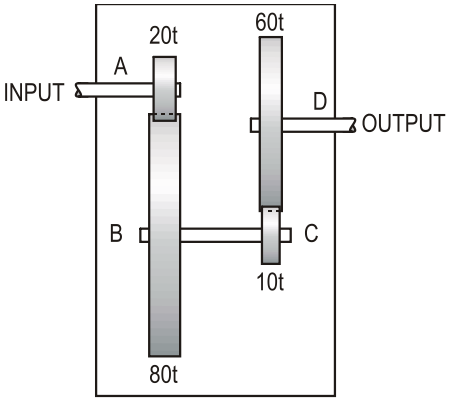


If gears are required to produce a very large change in speed, for example 100:1 then problems can arise with the size of gear wheels if a simple gear train is used. The problem can be overcome by mounting pairs of gears on the same shaft as shown.

This diagram shows how the shafts are connected between the 'pairs' of gears. Gears B and C are connected and rotate at the same speed. To calculate the **Velocity Ratio** for the gear train, it is necessary to calculate the ratio for each pair of meshing gears.

Worked teacher example:

Consider the following compound gear train shown opposite.



If the driver rotates at 100 rev/min in a clockwise direction, calculate…

a) The Velocity Ratio

b) The output speed of the system

c) The direction of rotation of the driven gear

Answers:

a) The velocity ratio for the first pair of meshing teeth is:

Compound gear train: Complete the following tasks in your jotter

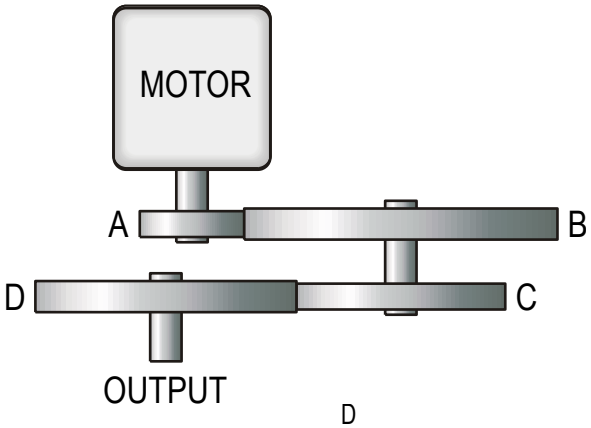
25. A compound gear train is shown below and is driven by a motor that runs at 1000 rev/min.

A = 20 teeth

B = 60 teeth

C = 40 teeth

D = 50 teeth



Calculate;

(a) The velocity ratio of the system.

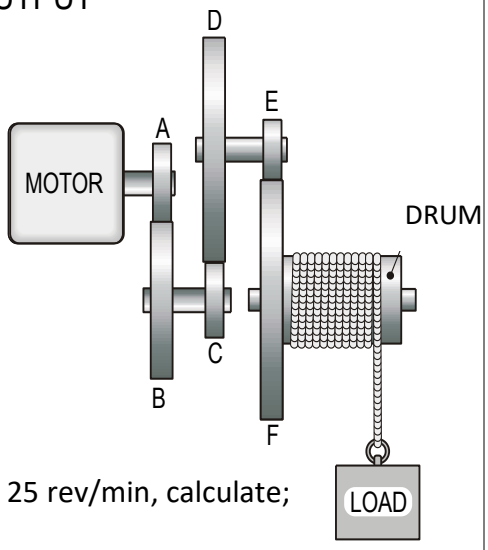
(b) The output speed.

26. A motorised winch is shown opposite.

A = ?

B = 32 teeth, C = 15 teeth, D = 45 teeth,

E = 12 teeth, F = 48 teeth



If the winch runs at a speed of 1200 rev/min and the drum is to rotate at 25 rev/min, calculate;

(a) The velocity required to produce this speed reduction.

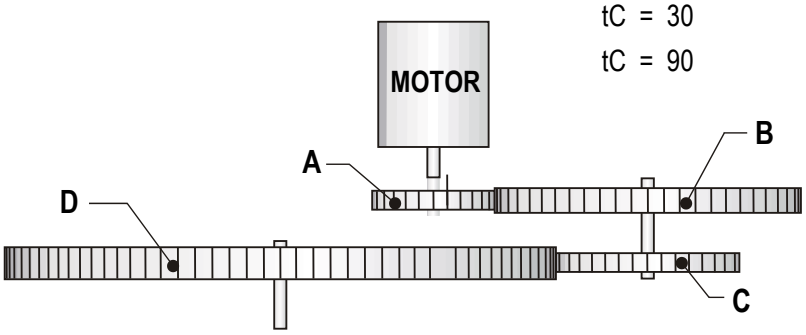
(b) The number of teeth gear A must have to meet this requirement.

27. The compound gear train shown is driven by a motor that runs at 750 rev/min. Calculate;

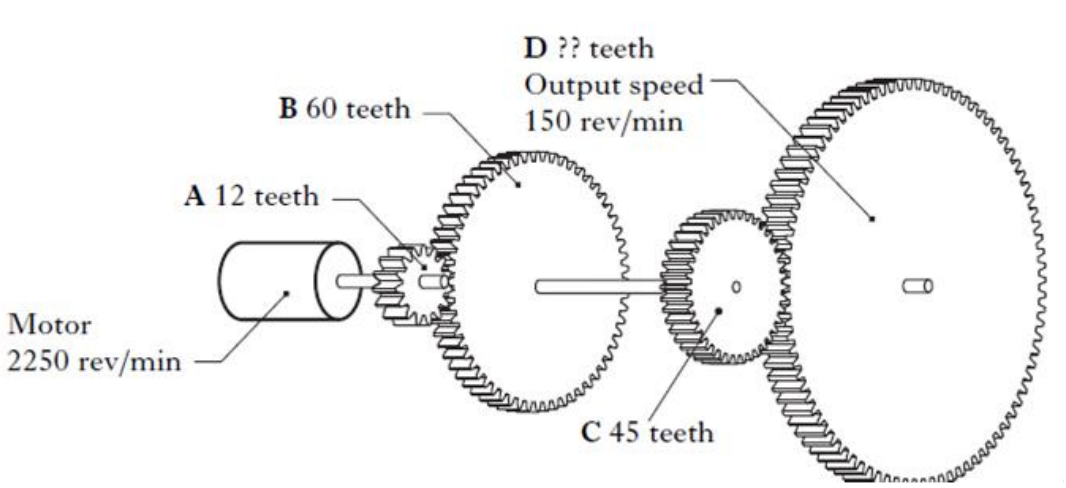
(a) The velocity ratio.

(b) The output shaft speed.

tA = 20
tB = 50
tC = 30
tC = 90



28. A gear system for operating a theatre curtain is shown.



(*a*) State the name of the type of gear system.

(*b*) Calculate:

(i) The velocity ratio of the system when gear D rotates at 150 rev/min.

(ii) The ratio of gear B to gear A;

(iii) The number of teeth on gear D to produce the correct output speed.

(*c*) State **one** advantage of a gear system made from plastic rather than steel gears.

29. A winch is to be designed for the front of an off-road vehicle. Use ***Yenka*** to simulate.



a) A simple gear train where the output turns 5 times slower than the input. (Use a constant speed motor as an input)

b) A compound gear train where the output turns 20 times slower than the input. (Use a constant speed motor as an input)

Draw your solution using the correct symbols for this compound gear train.

30. A motor with a single worm wheel output rotates at 500 rev/min. You are given the following sizes of worm gears which to select.

a) 10 teeth
b) 25 teeth
c) 50 teeth

Explain which gear should be connected to the motor to give the slowest output speed and why? Prove the system on Yenka.

All further questions should be attempted within your jotter.

31. A pedal cycle has 60 teeth on the driver sprocket and 10 teeth on the driven sprocket. What is the velocity ratio of the chain-drive system?

32. The bicycle shown has 2 rear sprockets, one with 50 teeth and the other with 80 teeth. The driver sprocket has 200 teeth. Calculate the output torque for the two rear sprockets if the input torque is 20 Nm.



33. During the construction of bridges and other tall structures, concrete is often pumped up to a working level. A motor can be used to power the pump but requires gear systems to ensure the pump moves at the correct speed.

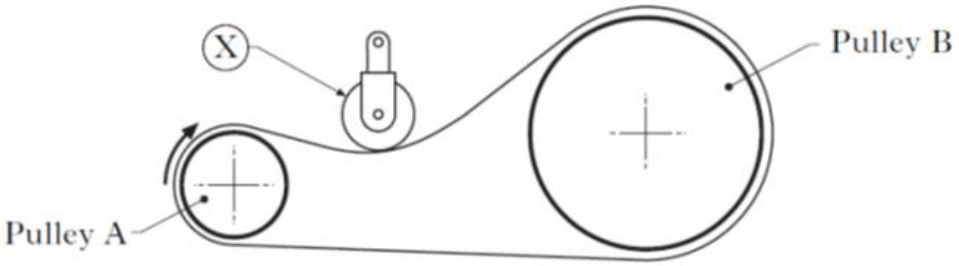
a) Draw, using appropriate symbols, a simple gear train that includes a 24 tooth driver and a 12 tooth driven.

b) Draw, using appropriate symbols, a compound gear system that includes two pairs of gears, and achieves the lowest possible ratio. The gear sizes available are 40, 24, 16 and 8 teeth.

c) Simulate both mechanisms using Yenka. Take a print screen .

Exam style questions

An air compressor on a heavy goods vehicle is driven by a belt.



(*a*) State the name of device (X) in the sketch above.

_____

Pulleys can rotate either **clockwise** or **anti-clockwise**.

Pulley A rotates clockwise, as shown in the diagram above.

(*b*) State the direction of rotation of device (X) .

_____

Three different types of belts are shown below.

(*c*) State the name of each belt.



(i) _____ (ii) _____ (iii) _____

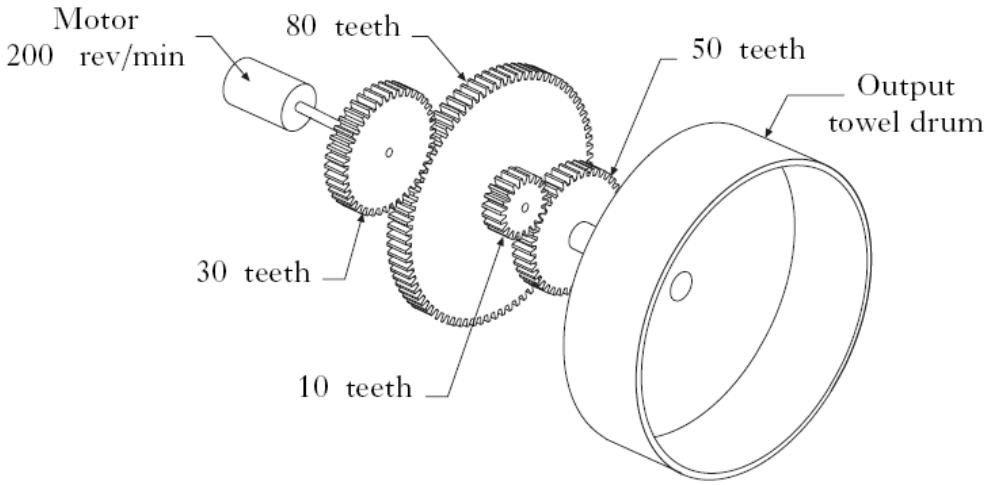
Friction is the resistance of one surface to move over another.

(*d*) State **two** methods of **reducing** friction.

1 _____

2 _____

A motorised towel dispenser uses a compound gear train as shown.

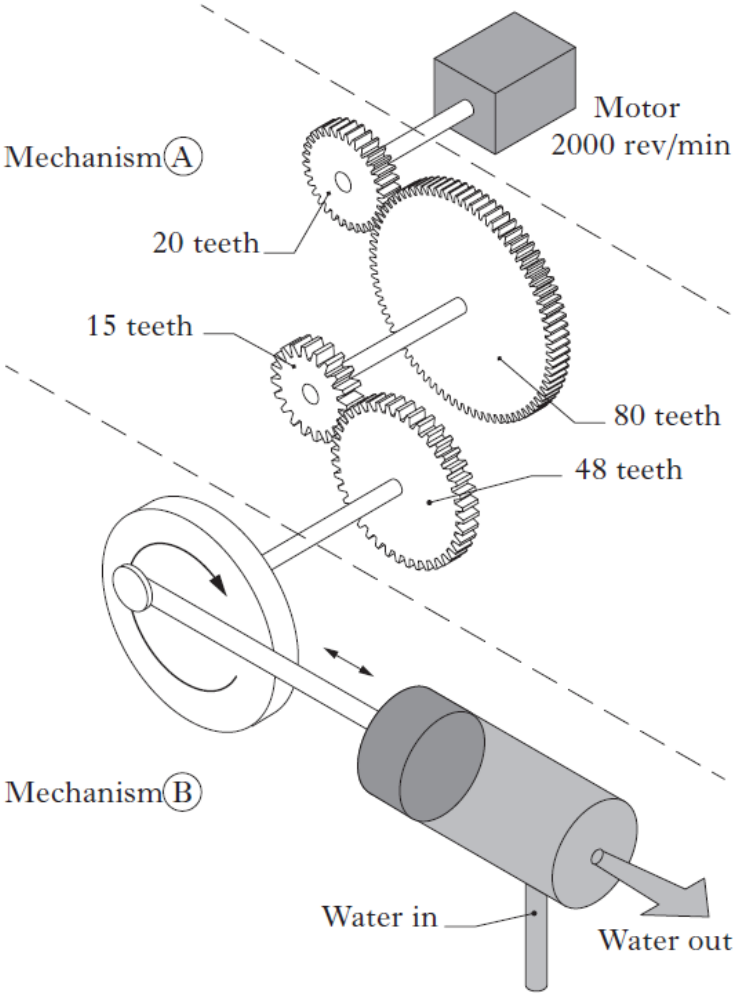


a) Calculate the multiplier ratio for the gear system.

b) Describe the difference between the input and output speed.

c) Explain the difference between the input and output direction produced by this drive mechanism.

A solar-powered water-pumping system is being tested for use in developing countries.



The system consists of **two** separate mechanisms.

(*a*) State the name of the following mechanisms.

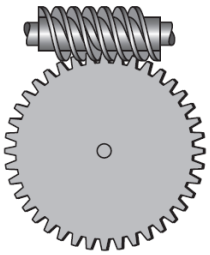
Mechanism (A) _____

Mechanism (B) _____

(*b*) Describe the change in motion produced by mechanism (B).

_____

Cont'd. Mechanism A is decided to be too bulky and is replaced by the following mechanism.



(*d*) State the name of this mechanism.

_____

(*e*) Describe how friction could be reduced in a mechanical system.

_____

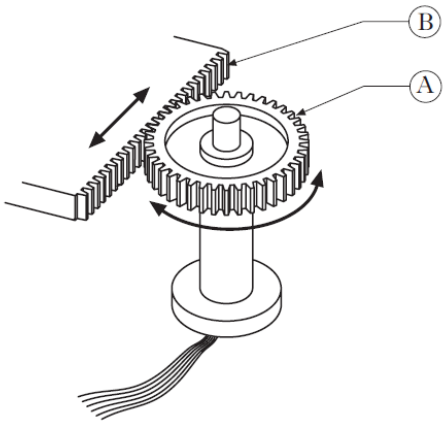
_____

(*f*) State the names of **two** mechanisms that will convert rotational motion into linear motion.

1 _____

2 _____

A DVD is ejected froma games console using the gear system shown.



(*a*) State the name of the parts of the gear system.

Ⓐ _____

Ⓑ _____

(*b*) State the **type** of input **motion** and output **motion** of the gear system shown above.

Input **Motion** _____

Output **Motion** _____